

CS4031 – Compiler Construction

Course Project Proposal

Spring 2026

Project Title: [e.g., "Chip-8 Coder", "AdventureScript", "PixelLang"]

Theme: [e.g., Chip-8 Inspired / Turtle Graphics DSL / etc.]

Team Members:

Student Name	Student ID
1. [Name]	[ID]
2. [Name]	[ID]
3. [Name]	[ID]

1. Language Concept

(A brief, 3-4 sentence description of your language. What is its purpose? Who is it for?)

Example: *"Chip-8 Coder" is a high-level language designed to simplify the creation of programs for the Chip-8 virtual machine. It will provide simple control structures and high-level operations that compile directly to Chip-8 assembly, making retro game development more accessible.*

2. Key Features

(List 3-5 core features your language will support.)

- Feature 1: [e.g., Support for 16 general-purpose 8-bit variables (V0-VF)]
- Feature 2: [e.g., Clear screen and draw-sprite commands]
- Feature 3: [e.g., If-else statements and simple loops]
- Feature 4: [e.g., Built-in functions for handling timer and keypad input]
- Feature 5: [e.g., Constants and basic arithmetic expressions]

3. Example Program

(Provide at least one small but concrete code snippet showing what your language looks like.)

text

```
// A simple program in our language
```

```
x = 10
```

```
y = 5
```

```
draw_sprite x, y, my_sprite_data
```

```
if key == 0xA then
```

```
    x = x + 1
```

```
end
```

```
loop 10 times
```

```
    clear_screen
```

```
    draw_sprite x, y, my_sprite_data
```

```
    delay 100
```

```
end
```

4. Target Output

(What does your compiler produce?)

- Assembly code (e.g., for Chip-8, a custom CPU)
- Bytecode for a custom Virtual Machine
- An interpreted language (like Python or JavaScript)
- A data file (e.g., image, sound, game map)
- Direct execution and display of results
- Other: [Please Specify]

5. Expected Challenges

(Identify 2-3 potential technical or design difficulties.)

- Challenge 1: [e.g., Correctly translating high-level loops into Chip-8's limited branching instructions.]
- Challenge 2: [e.g., Designing a robust symbol table to handle variable scoping.]
- Challenge 3: [e.g., Generating efficient code for sprite drawing operations.]